

LLM 2.2.

LLM lar chiqti degani bu dastur chilar
kerak emas degani emas.

Productni qurish osson lekin ularni
manage qilish asosiy varifalardan biri ekan

.gitignore → private saqlashimiz kerak bo'lgan
va kotta hajmdagi ma'lumotlarni
git hubga push qilmaslik u-n kerak.
Ex. .cache, checkpoints, weights, vram, huggingface.

Token bu ishlagand privacy ga e'tiborki bolish
kerak ekan.

ML vs LLM RISKS.

ML → data structured bo'lgani u-n hajm
jihatidan kotta bo'lmaydi va natijada
model size ham kotta bo'lmaydi.
Datasetlar xavf tugatmaydi.

Dataset kichik va RW va Transformerlar
bölmağari uchun risk va model re-training
juda soada va aʼron bōladi.

LLM → LLM da model parameter va weightlari
kōpligi sababli juda katta hajmga
bōladi va bu Git u-n tōsiq bōladi

Datasetlar qimmatli bōladi va asosan
text xolatida bōladi va hajm jixatdan
juda kam kotta ekan.

Data Sources.

LLM train qilayotganda qaysi yōnalish
uchun data tanlash juda muhim.

Asosiy sources:

- Hugging face / fine web
- Common Crawl
- Dolma

Note: Har bir yōnalish u-n common data
source mavjud: Fine web, FineWeb-Edu,
The Pile, Red Pajama, Dolma, Stack v2, azXiv.

1. Dataset

FinWeb

FinWeb Edu

The Stack

2. Download Small Subset

3. Clean Data

4. Deduplicate

5. Tokenization

6. Token Dataset

7. Train SMM

8. Evaluate

9. Fine-Tuning

10. Deploy

Quality Assurance: ~~Dimension~~

Yaxshi model qurish uchun: qarat kotta dataset kerak degani emas.

1. Relevance

2. Quality Grammar / Logic

3. Diversity, Turli xil datayigish / domeinda.

4. Accuracy / Factual Quality Ishonsa Bolorlimi ???

5. Safety → Hate speech / terrorism
6. Duplication → Takrorlangan data ni tozalash
7. Sensitivity → Shaxsiy data

Data Mixture:

Ha kitob o'rgananolan kora turki xil
kitob o'zigan yaxshi.

license checklist:

- What is license?
- Owner
- Allowed actions
- Commercial allowed??
- Redistribution allowed?
- ~~Dis~~ Attribution required
- Any additional restrict's.

yamf filga qaysi 'license' ekanligini
yozib qo'shish kerak

ODC-by and MIT → all good / commercial &

CC BY-NC-ND → No Commercial / Modif, redistri-
bution. → Depends

CC-BY-NC, CC BY-NC-SA → No commercial,
Modifg → Yes, Redistribution-Dep